

Definition of Sample Covariance. The **sample covariance** (s_{XY}) is a measure of how two variables in a sample move together:

$$s_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}. \quad (10)$$

Equation 10 indicates that the sample covariance is the average value of the product of the deviations of observations on two random variables (X_i and Y_i) from their sample means. If the random variables are returns, the units would be returns squared. Also, note the use of $n - 1$ in the denominator, which ensures that the sample covariance is an unbiased estimate of population covariance.

Stated simply, covariance is a measure of the joint variability of two random variables. If the random variables vary in the same direction—for example, X tends to be above its mean when Y is above its mean, and X tends to be below its mean when Y is below its mean—then their covariance is positive. If the variables vary in the opposite direction relative to their respective means, then their covariance is negative.

The size of the covariance measure alone is difficult to interpret as it involves squared units of measure and so depends on the magnitude of the variables. This brings us to the normalized version of covariance, which is the correlation coefficient.

Definition of Sample Correlation Coefficient. The **sample correlation coefficient** is a standardized measure of how two variables in a sample move together. The sample correlation coefficient (r_{XY}) is the ratio of the sample covariance to the product of the two variables' standard deviations:

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}. \quad (11)$$

Importantly, the correlation coefficient expresses the strength of the linear relationship between the two random variables.

Properties of Correlation

We now discuss the correlation coefficient, or simply correlation, and its properties in more detail:

1. Correlation ranges from -1 and $+1$ for two random variables, X and Y :
 $-1 \leq r_{XY} \leq +1$.
2. A correlation of 0 , termed uncorrelated, indicates an absence of any linear relationship between the variables.
3. A positive correlation close to $+1$ indicates a strong positive linear relationship. A correlation of 1 indicates a perfect linear relationship.
4. A negative correlation close to -1 indicates a strong negative (i.e., inverse) linear relationship. A correlation of -1 indicates a perfect inverse linear relationship.

We return to scatter plots to illustrate correlation visually. In contrast to the correlation coefficient, which expresses the relationship between two data series using a single number, a scatter plot depicts the relationship graphically. Therefore, scatter plots are a very useful tool for the sensible interpretation of a correlation coefficient.

Exhibit 26 shows examples of scatter plots. Panel A shows the scatter plot of two variables with a correlation of $+1$. Note that all the points on the scatter plot in Panel A lie on a straight line with a positive slope. Whenever variable X increases by one unit, variable Y increases by two units. Because all of the points in the graph lie on a straight line, an increase of one unit in X is associated with an exact two-unit increase in Y , regardless of the level of X . Even if the slope of the line were different